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Assignment 2: Perform the following operations using Python on the data sets:

1. Compute and display summary statistics for each feature available in the dataset. (e.g. minimum value, maximum value, mean, range, standard deviation, variance and percentiles
2. Illustrate the feature distributions using histogram.
3. Data cleaning, Data integration, Data transformation, Data model building (e.g. Classification)

Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the dataset

```
train = pd.read_csv('/content/train.csv')
test = pd.read_csv('/content/test.csv')

train.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
				Futrelle, Mrs. Jacques								

Check Shape of Dataset

```
train.shape
```

```
(891, 12)
```

Summary Statistics

```
train.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Range of each feature

```
train.max(numeric_only=True) - train.min(numeric_only=True)
```

	0
PassengerId	890.0000
Survived	1.0000
Pclass	2.0000
Age	79.5800
SibSp	8.0000
Parch	6.0000
Fare	512.3292

dtype: float64

Variance

```
train.var(numeric_only=True)
```

	0
PassengerId	66231.000000
Survived	0.236772
Pclass	0.699015
Age	211.019125
SibSp	1.216043
Parch	0.649728
Fare	2469.436846

dtype: float64

Standard Deviation

```
train.std(numeric_only=True)
```

	0
PassengerId	257.353842
Survived	0.486592
Pclass	0.836071
Age	14.526497
SibSp	1.102743
Parch	0.806057
Fare	49.693429

dtype: float64

Percentiles

```
train.select_dtypes(include=np.number).quantile([0.25, 0.5, 0.75])
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
0.25	223.5	0.0	2.0	20.125	0.0	0.0	7.9104
0.50	446.0	0.0	3.0	28.000	0.0	0.0	14.4542
0.75	668.5	1.0	3.0	38.000	1.0	0.0	31.0000

Data Types

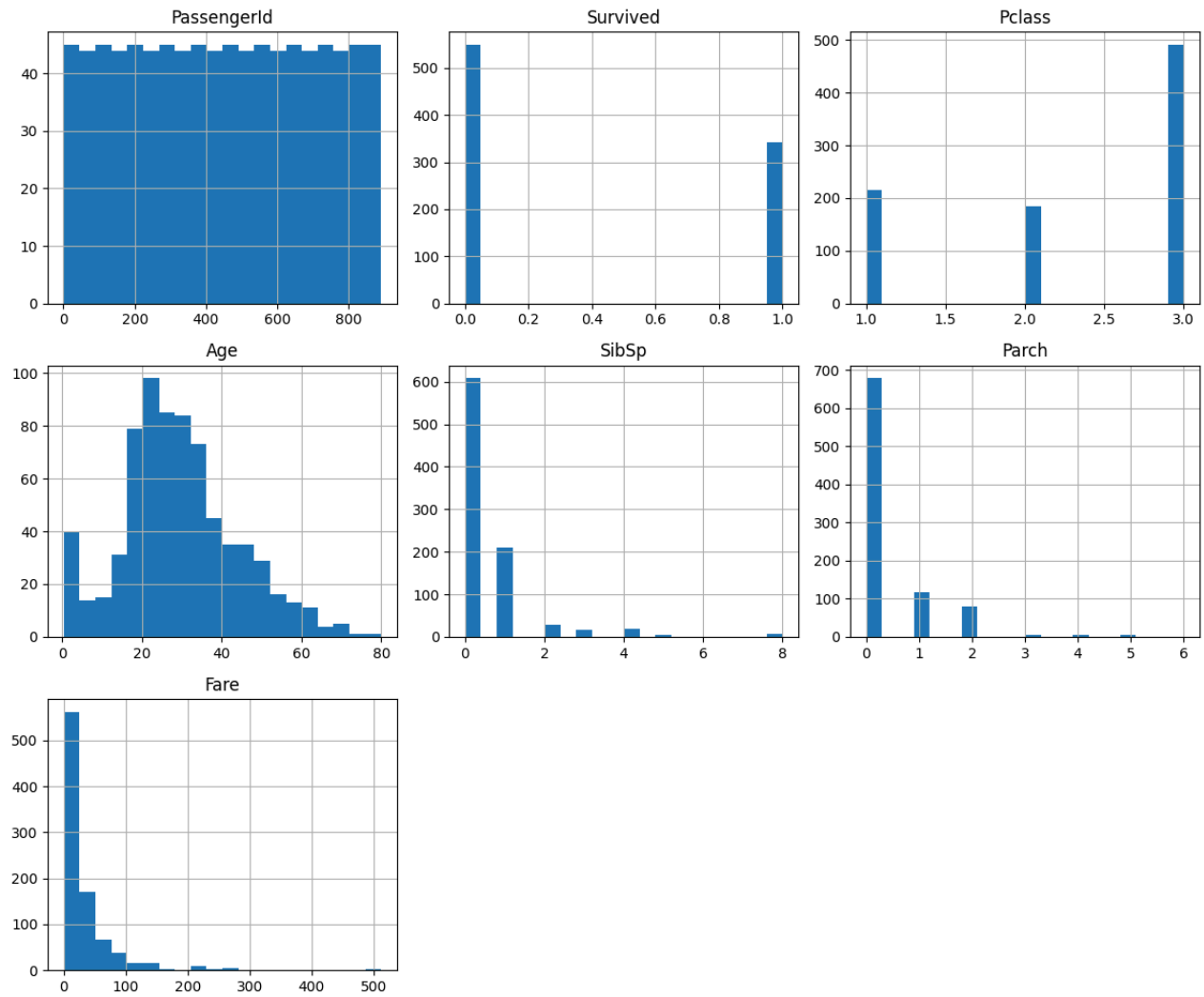
```
train.dtypes
```

	0
PassengerId	int64
Survived	int64
Pclass	int64
Name	object
Sex	object
Age	float64
SibSp	int64
Parch	int64
Ticket	object
Fare	float64
Cabin	object
Embarked	object

dtype: object

Histograms for Feature Distribution

```
train.hist(figsize=(12,10), bins=20)
plt.tight_layout()
plt.show()
```



Data Cleaning

```
train.fillna(value=5)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	5	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	5	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	5	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	5	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	5.0	1	2	W./C. 6607	23.4500	5	S

Find Missing values

```
train['Age'] = train['Age'].fillna(train['Age'].median())
train['Embarked'] = train['Embarked'].fillna(train['Embarked'].mode()[0])
train['Cabin'] = train['Cabin'].fillna('Unknown')
test['Age'] = test['Age'].fillna(test['Age'].median())
test['Fare'] = test['Fare'].fillna(test['Fare'].median())
test['Cabin'] = test['Cabin'].fillna('Unknown')
```

Remove Duplicates

```
train = train.drop_duplicates()
```

```
train.dropna(how="any")
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	Unknown	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	Unknown	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	Unknown	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	Unknown	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S

```
pd.isna(train)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...	...
886	False	False	False	False	False	False	False	False	False	False	False	False
887	False	False	False	False	False	False	False	False	False	False	False	False
888	False	False	False	False	False	False	False	False	False	False	False	False
889	False	False	False	False	False	False	False	False	False	False	False	False
890	False	False	False	False	False	False	False	False	False	False	False	False

891 rows × 12 columns

Data Integration (train + test)

```
combined = pd.concat([train, test], axis=0, ignore_index=True)
combined.head()
```

Data Transformation

1. Drop Irrelevant Columns
2. Encode Categorical Variables

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0.0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	Unknown	S
1	2	1.0	1	Cumings, Mrs. John Bradley (Florence	female	38.0	1	0	PC 17599	71.2833	C85	C

```
columns_to_drop = ['PassengerId','Name','Ticket','Cabin']
train = train.drop(columns=[col for col in columns_to_drop if col in train.columns], axis=1)
test = test.drop(columns=[col for col in columns_to_drop if col in test.columns], axis=1)
train = pd.get_dummies(train, drop_first=True)
test = pd.get_dummies(test, drop_first=True)
```

Concat

```
import pandas as pd

academic1 = pd.DataFrame({
    "Student_ID": [101, 102, 103],
    "Name": ["Nisha", "Sohaa", "Apurv"],
    "Marks": [85, 92, 78]
})

academic2 = pd.DataFrame({
    "Student_ID": [104, 105],
    "Name": ["Priya", "Karan"],
    "Marks": [88, 95]
})

attendance = pd.DataFrame({
    "Student_ID": [101, 102, 103],
    "Attendance (%)": [90, 95, 85]
})
```

Vertical Merge

```
vertical_merge = pd.concat([academic1, academic2], axis=0)
print(vertical_merge)
print("\n")
```

```
   Student_ID   Name  Marks
0         101  Nisha     85
1         102  Sohaa     92
2         103  Apurv     78
0         104  Priya     88
1         105  Karan     95
```

Horizontal Merge

```
horizontal_merge = pd.concat([academic1, attendance], axis=1)
print(horizontal_merge)
```

```
   Student_ID   Name  Marks  Student_ID  Attendance (%)
0         101  Nisha     85         101              90
1         102  Sohaa     92         102              95
2         103  Apurv     78         103              85
```

Merge

```
academic = pd.DataFrame({
    "Student_ID": [101, 102, 103],
    "Name": ["Nisha", "Sohaa", "Apurv"],
    "Marks": [85, 92, 78]
})
attendance = pd.DataFrame({
    "Student_ID": [101, 102],
    "Attendance (%)": [90, 95]
})
inner_join = academic.merge(attendance, on="Student_ID", how="inner")
print(inner_join, "\n")
```

```
   Student_ID   Name  Marks  Attendance (%)
0         101  Nisha     85              90
1         102  Sohaa     92              95
```